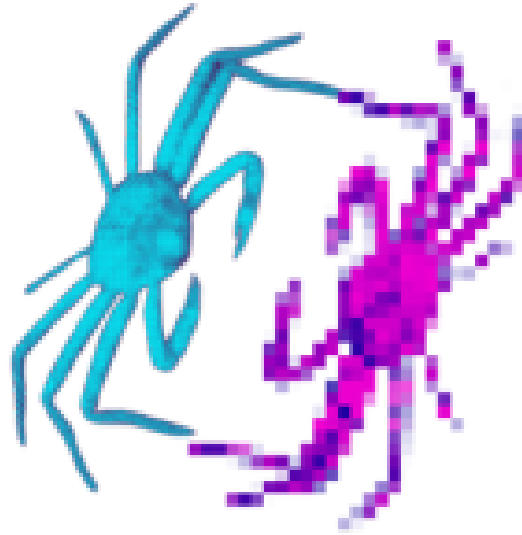




Computer Vision

Across the Marine Sciences

Instructor Manual

ocean cv.org

A teaching companion to the open textbook and Colab notebooks.

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Part 1: Course Overview

IM-1 About This Manual

The OceanCV textbook is a free, open, interactive resource for learning computer vision in the context of marine biology, fisheries, and oceanography. Every chapter pairs conceptual material with a runnable Google Colab notebook, so learners read and run code at the same time. The textbook is structured so that someone working alone can work through it at their own pace, but it also works well as the backbone of a taught course.

This manual is the teaching companion. It is written for two audiences. The first is instructors who want to adapt the notebooks and text into a course or workshop of their own. The second is independent learners who want concrete guidance on how to sequence the material. It is not a script and it does not tell you word for word what to say. It is a set of practical suggestions, informed by three offerings of the course at the University of Washington.

A few things this manual does not try to do. It does not reproduce the textbook content. It does not cover how to write Python. It assumes you have read the chapters you are teaching. What it does give you is timing estimates, notes on common student difficulties, adaptation guidance for different formats, and a per-lesson reference you can scan before class.

IM-2 What the Course Covers

Students learn to work with marine image data end to end. They annotate images, build and train models, evaluate performance, and interpret results critically. The core tools are Python, PyTorch, Ultralytics YOLO (version 26), and TensorFlow. Problems covered include species detection, benthic cover estimation, keypoint-based morphometrics, super-resolution, and satellite change detection.

The course is not about memorizing API calls. It is about reading model outputs and deciding what the model is actually doing. A student who finishes the course should be able to look at a precision-recall curve and say something meaningful about whether the model is good enough for a specific use case, and why.

IM-3 Teaching Philosophy

The course is intentionally low-code and interpretation-first. Most class time goes toward reading and reasoning about model outputs: training curves, confusion matrices, detection overlays on real imagery. Students spend less time writing code from scratch and more time asking what the output actually means for the scientific question.

Sessions are hands-on and studio-like. Students work on laptops or lab computers, usually in pairs or small groups, running notebooks in Google Colab while the instructor circulates and helps. This format works at small workshop scale and at lecture-course scale. The notebooks require no local GPU and run on Colab's free

tier, so the approach scales from a single laptop up to a department compute cluster without changing anything in the notebooks.

Instructor note

The core skill is interpretation, not coding. Students who leave able to read a confusion matrix and a precision-recall curve, and who know when to trust a model, have the main skill even if they never write a training loop from scratch.

IM-4 How the Materials Are Organized

The course has four main chapters plus a chapter on Vision-Language Models and several appendices.

Chapter 1 (Introduction to Marine Imaging) provides context. It covers why marine imaging is hard, what kinds of image data exist in oceanographic and fisheries science, and what a neural network is doing at a conceptual level. Lessons include an introduction to marine imaging, imagery fundamentals, ocean image types, introduction to AI, transfer learning, and model interpretability.

Chapter 2 (Introduction to Computer Vision) covers the data side. Students learn to annotate images, explore and manipulate datasets, apply augmentation, and read the metrics used to evaluate CV models. Lessons include image annotation for CV, manual annotation, ImShow data exploration, image manipulation with PIL and OpenCV, creating and augmenting datasets, the augmentation Streamlit app, and understanding CV metrics.

Chapter 3 (Computer Vision Development) is the core technical chapter. It covers image classification, object detection, keypoint detection, instance segmentation, classical segmentation, super-resolution, GANs, model interpretability in practice, edge deployment, live inference, and benthic semantic segmentation. Lessons include image classification with Keras, introduction to YOLO, training and deploying object detection with YOLO, ResNet and Keras detection alternatives, ice seal classification, OceanCV localization, keypoint detection, instance segmentation, K-means segmentation, MarineInst segmentation, SRGAN super-resolution, Sea Star GAN, whale shark interpretability, edge deployment optimization, live inference with PyQt, and benthic semantic segmentation.

Chapter 4 (Synthesis Project) walks students through the final project workflow. It covers dataset selection, Hugging Face organization setup, model card writing, Streamlit apps, and Gradio apps.

The VLM Chapter (Vision-Language Models) covers the current generation of multimodal models. It is optional and marked as cutting edge. Lessons include a VLM introduction, MLP projection, Q-Former, native multimodal architectures, the Perceiver Resampler, and high-resolution grounding.

Appendices include a bibliography, a pre-course survey, and a lesson covering the math behind convolutional neural networks.

IM-5 Audience and Prerequisites

The course assumes basic Python syntax, roughly what someone would have after a single introductory programming or scientific computing course. No machine learning or computer vision background is assumed or required. Marine biology knowledge is useful context but not a prerequisite.

Most students find that the hardest adjustment is not the code. It is getting comfortable with the idea that model outputs are probabilistic and imperfect. That adjustment is itself a useful scientific skill, and it tends to happen gradually over the first few weeks.

IM-6 Technology and Compute

Learners need three accounts. First, a Google account for Colab. Second, a free Hugging Face account with a request to join the [OceanCV Organization](#) hub. Third, an account on a labeling platform, either CVAT or Label Studio. No local GPU is required.

Course adopters setting up their own course or laboratory hub should refer to the official [Hugging Face Organizations Documentation](#) for step-by-step instructions on creating organizations, managing member roles, and establishing shared repositories.

A note for course adopters on GPU access. Training the YOLO models in Chapter 3 works fine on Colab's free GPU for small datasets and short runs, but free-tier GPU availability is unpredictable and sessions have time limits. For larger classes or longer training runs, institutional Colab Pro licenses or a shared JupyterHub with GPU access are worth the cost. The localization and interpretation lessons can run on CPU if needed, since they use a pretrained model.

IM-7 Assessment Model

The course uses four assessment categories and no final exam.

Category	Weight	Notes
Reading quizzes / Homework	15%	Short autograded 5-question checks. Lowest score dropped.
In-class activities	15%	Paper worksheets with graph interpretation and short-answer prompts. Lowest score dropped.
Exit tickets	10%	Brief end-of-class survey. Lowest score dropped.
Synthesis project	60%	Proposal presentation (20%), model card (20%), final presentation (20%). No drop.

Instructor note

The synthesis project is the heart of the course. Keep it in some form even when compressing to a shorter format. The other categories are swappable.

IM-8 The Final Project

The synthesis project is the capstone. Each student or pair finds a marine image dataset, trains or localizes a model, and presents their results. Grading focuses on documentation, scientific reasoning, and honest interpretation of results. Accuracy metrics are not the target. A model that fails in an interesting way, with a student who can explain why, earns a better grade than a black-box success with no analysis.

Integrate the project early. Here is the recommended timeline.

1. **Day 1.** Ask what problems students might want to work on. Plant the idea early.
2. **Week 2.** Run a short brainstorm of problems they could realistically solve with the tools covered so far.
3. **Weeks 3-4.** Have students browse dataset repositories while they are actively learning detection. Looking at real data motivates the technical material.
4. **Week 6.** One-page written proposal naming the dataset, the CV task, and the model type they plan to use.
5. **Week 7.** Proposal presentations. Short, informal, five minutes each.

IM-9 Adapting the Materials

Chapters are modular. Each has an overview, learning objectives, worked examples, and exercises. You can reorder them or drop them without breaking later material, with one exception. The synthesis project chapters (22-25) assume that students have completed at least one detection or segmentation lesson, since the project requires training or localizing a model. Everything else is fair game to swap.

Part 2: Suggested Timelines

IM-10 Core vs Optional Chapters

Not every chapter needs to be taught. The table below marks chapters as core or optional. Core chapters form a coherent, complete course on their own. Optional chapters add depth, alternate frameworks, or coverage of advanced topics. For a short course or workshop, start by cutting optional chapters and protect the core.

Chapter	Topic	Role
Getting Started	Setup, accounts, environments	Core – do first
1, 3, 4	Marine imaging, ocean image data, AI in marine science	Core – context
7, 8	Image annotation for CV; manual annotation	Core
11	Creating and augmenting datasets	Core
12	Understanding CV metrics and graphs	Core – spend the full session here
14, 15	YOLO26 tasks and modes; training and deploying object detection	Core
17	Localizing the OceanCV FirstPass model to a new dataset	Core
19	Instance segmentation with YOLO26	Core
22–25	Synthesis project: datasets, HF org, model card, Streamlit	Core – capstone
2	Imagery fundamentals	Optional – good homework if new to digital imaging
5, 6	Transfer learning; model interpretability	Optional – conceptual depth
9, 10	Exploring data with ImShow; PIL/OpenCV manipulation	Optional – Python fluency
13, 16	Classification with Keras; ice seal worked example	Optional – alternate framework
18	Keypoint detection with YOLO26	Optional – extension task
20	K-means segmentation	Optional – classical contrast
21	SRGAN super-resolution	Optional – advanced
VLM lessons	Vision-language models	Optional – cutting edge
Appendix	The math behind CNNs	Optional – for math-inclined learners

IM-10.1 Quarter (10 weeks, as taught)

This is the format the course has been delivered in at the University of Washington. It is tight. Weeks 3 and 4 carry the most technical load. If something has to slip, the optional chapters in week 5 are the easiest to compress.

Week	In-class focus	Assigned before class
1	Course setup; introduction to marine imaging and ocean image data	Getting Started; Ch 1, 3, 4
2	Annotation practice; introduction to CV, YOLO26, and object detection	Ch 7, 8, 14
3	Interpreting model metrics; reading training graphs	Ch 12
4	Localizing an object-detection model to a new dataset	Ch 15, 17
5	Building from detection: tracking in zone, keypoint detection	Ch 16, 18
6	Instance segmentation and its variants	Ch 19, 20
7	Proposal presentations	Ch 22 (to prep proposal)
8	Project work time; Hugging Face model card workshop	Ch 23, 24
9	Project work time	Ch 25
10	Final presentations	—

IM-10.2 Semester (14-15 weeks, expanded)

A semester format gives time to cover transfer learning, model interpretability, and the optional Python fluency lessons. Students arrive at the technical chapters better prepared, and the final project has more breathing room.

Week	In-class focus	Assigned reading
1	Course intro and setup; marine imaging	GS, 1, 2
2	Ocean image data; AI in marine science	3, 4
3	Transfer learning; model interpretability	5, 6
4	Image annotation	7, 8
5	Exploring data; image manipulation with PIL and OpenCV	9, 10
6	Creating and augmenting datasets	11
7	Understanding CV metrics and graphs	12
8	Image classification with Keras; YOLO tasks and modes	13, 14
9	Object detection train and deploy; ice seal worked example	15, 16
10	Localizing FirstPass; keypoint detection	17, 18

Week	In-class focus	Assigned reading
11	Instance segmentation; K-means segmentation	19, 20
12	Super-resolution (optional); project dataset selection	21, 22
13	Proposal presentations; HF organization and model card	23, 24
14	Streamlit apps; project work time	25
15	Final presentations and course review	—

IM-10.3 Workshop (3-day intensive)

The workshop format compresses the core into three full days. Participants should arrive with their own marine imagery. Send the annotation guide (Chapter 7) at least one week in advance so they arrive ready to label. The end-of-day deliverable keeps each day concrete.

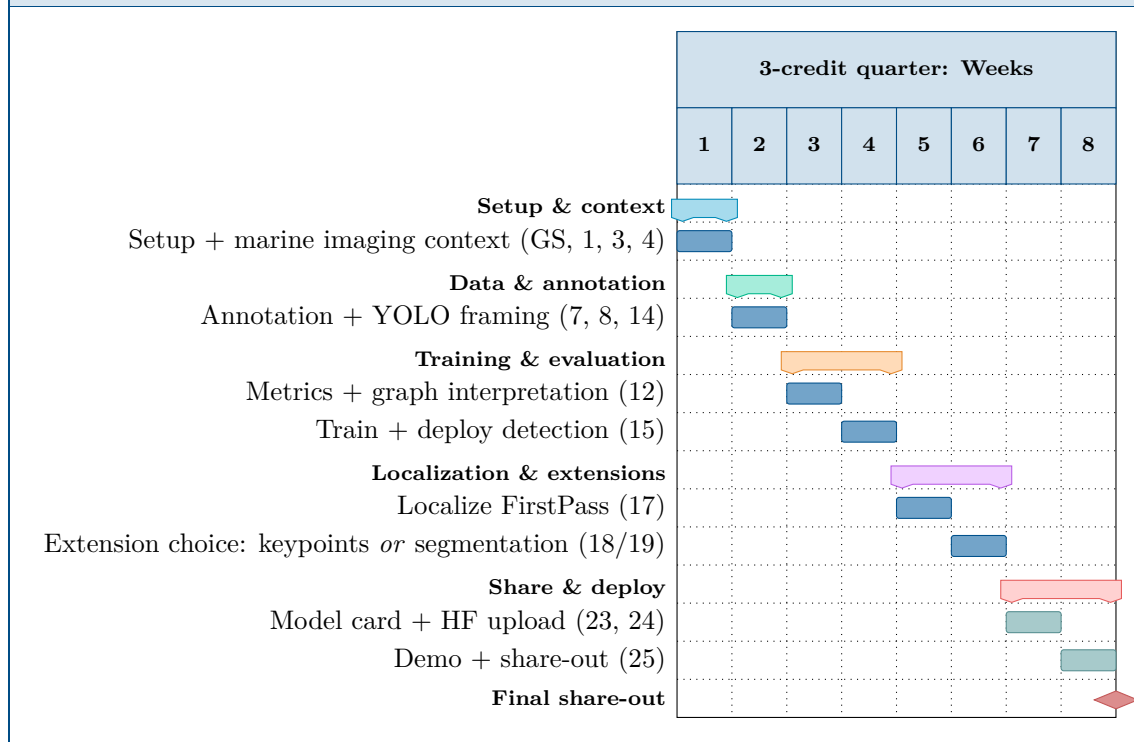
Day	Focus and activities	Chapters	End-of-day deliverable
1	Foundations and annotation: setup check, marine imaging context, annotation sprint, dataset augmentation	GS, 1, 3, 4, 7, 8, 11	An annotated, augmented dataset ready to train
2	Training and interpretation: YOLO tasks and modes, train a detection model, read metrics, localize FirstPass	14, 15, 12, 17	A trained detection model with a metrics read-out
3	Extend, deploy, and share: one extension task (instance segmentation or keypoints), model card, Hugging Face upload, Streamlit demo	19 or 18, 24, 23, 25	A model on Hugging Face with a model card and demo

IM-10.4 Quarter (8 weeks): 3-credit vs 5-credit

Legend	 In-session lesson	 Project / studio	 Milestone
Theme colors	 Context	 Data	
	 Training	 Extensions	
	 Share/demo		

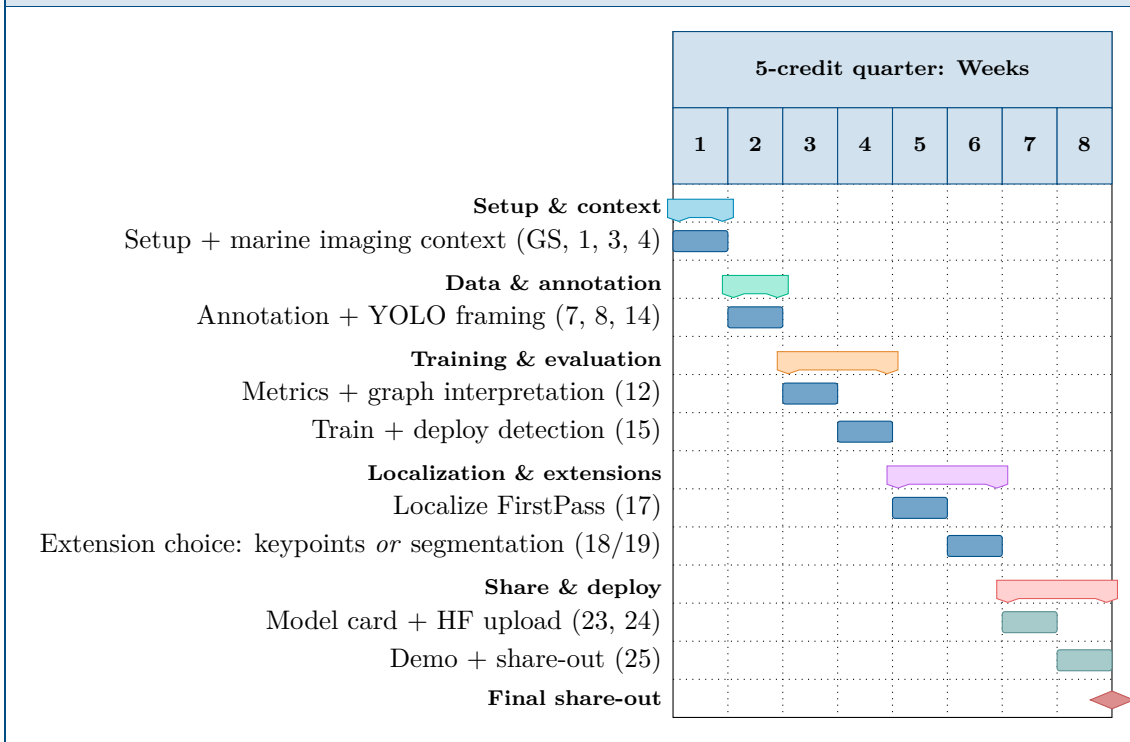
3-credit quarter (no code homework; reading only)

3-credit quarter (8 weeks) — core studio spine



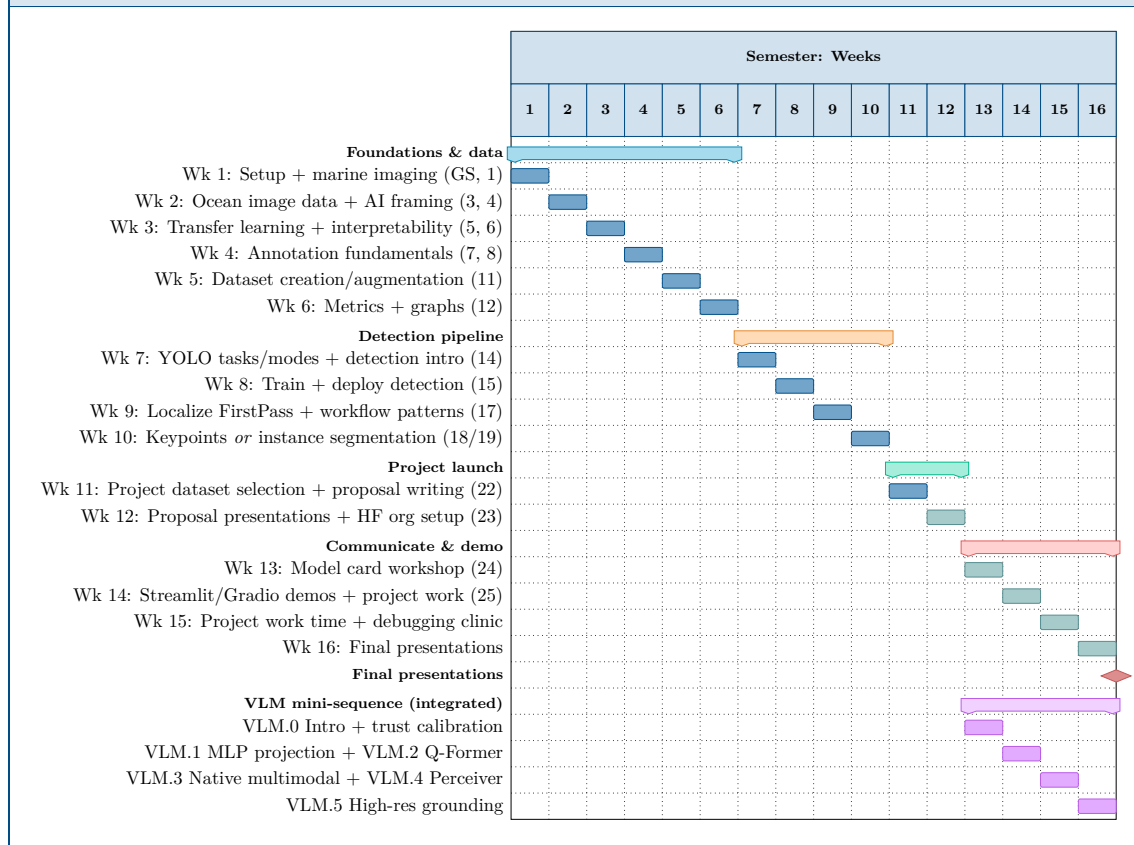
5-credit quarter (adds optional lessons as code homework)

5-credit quarter (8 weeks) — core spine (async prep not shown)



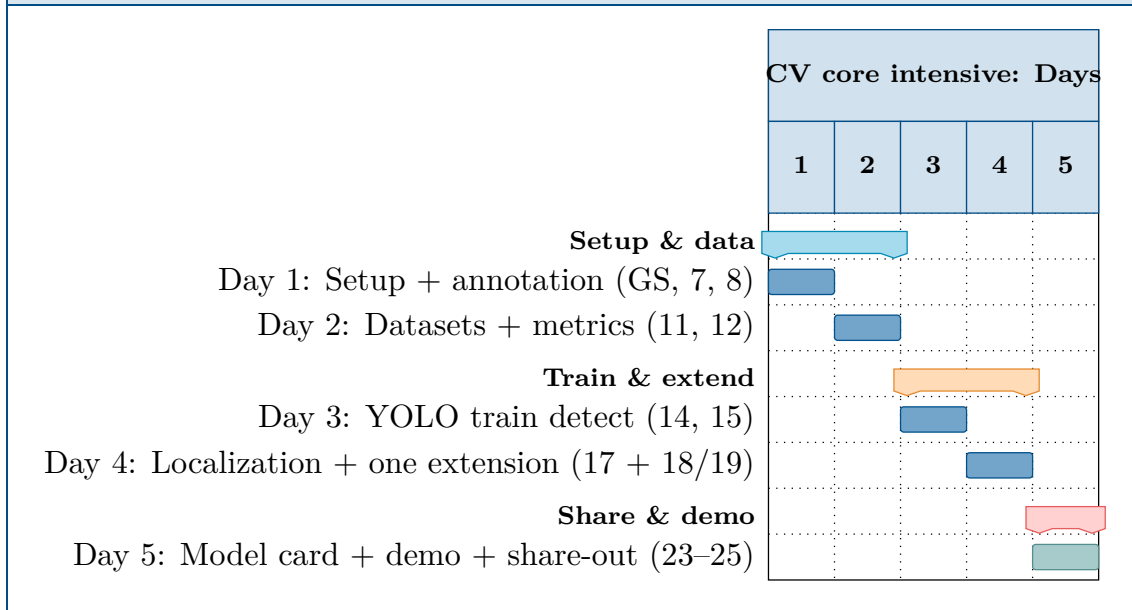
IM-10.5 Semester (16 weeks, full)

Semester (16 weeks) — full sequence



IM-10.6 Intensive (1 week): Computer Vision core

Intensive (1 week) — Computer Vision core (5 days)



IM-10.7 Intensive (1 week): Vision-Language Models (VLM)

Intensive (1 week) — Vision-Language Models (VLM) (5 days)

		VLM intensive: Days				
		1	2	3	4	5
Core sequence						
Day 1: VLM framing + trust calibration (VLM.0)						
Day 2: Architectures I (VLM.1, VLM.2)						
Day 3: Architectures II (VLM.3, VLM.4)						
Day 4: Grounding + structured outputs (VLM.5)						
Mini-project						
Day 5: Mini-project: prompt + evaluate on your imagery						

IM-10.8 Stand-alone module (1–2 days): Computer Vision

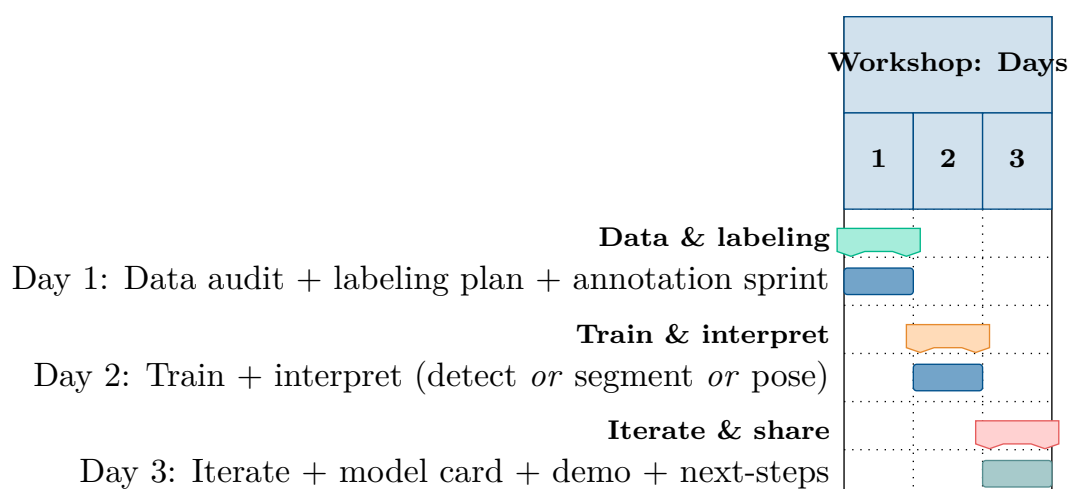
Day	In-session plan	Best for
Day 1	Rapid end-to-end: annotation sprint (7, 8) → train detect (14, 15) → read metrics (12).	Detection/counting use cases; learners new to CV.
Day 2 (optional)	Choose one extension: localization (17) <i>or</i> segmentation (19) <i>or</i> keypoints (18). Close with short model card (24).	Tailoring to domain: benthic cover, morphometrics, or cross-site transfer.

IM-10.9 Stand-alone module (1–2 days): Vision-Language Models (VLM)

Day	In-session plan	Best for
Day 1	Trust + failure modes (VLM.0) and practical prompting on marine imagery; introduce one architecture at a high level (VLM.1).	Qualitative triage, labeling assistance, rapid reporting.
Day 2 (optional)	Grounding + extraction (VLM.5), then evaluation: error taxonomy, hallucination checks, and a small benchmark on participant images.	Workflows where incorrect text is costly; structured downstream use.

IM-10.10 Workshop (3 days): Bring-your-own dataset (researcher-focused)

Workshop (3 days) — bring-your-own dataset



Instructor note

Pre-work (send 1 week before): bring 50–200 representative images, decide your target task (detect/segment/pose), and confirm you can legally use/share the data in class.

IM-10.11 Picking “extension tracks” by research interest

Research interest	What to emphasize	Suggested add-ons
Abundance / counting (fish, debris, megafauna)	Object detection + robust metrics interpretation.	15, 12, 17; optionally 16 (workflow patterns).
Benthic cover / habitat mapping	Segmentation and area-based metrics.	19; optionally 17 (cross-site/localization) and 20 (classical contrast).
Morphometrics (length/shape)	Keypoints + measurement pipeline.	18; optionally 12 (error analysis) and 11 (augmentation).
Deployment / field hardware	Export, speed/latency trade-offs.	15 (export basics) + 3.15 (Edge Deployment Optimization).
Rapid triage / weak labeling from text	VLMs as assistants, with evaluation and guardrails.	VLM.0 and VLM.5; pair with 7/8 if converting to labels.

IM-11 Adapting the Timelines

A few practical notes on customization.

- Swap optional chapters in or out to match learner background and available time.
- No GPU or no time to train? Use the pretrained FirstPass model and spend the saved time on localization and interpretation.
- Short on days in the workshop format? Drop the day-three extension task first, then trim the day-one context lessons. Protect the model card and demo.
- Pairs or small groups work better than solo for the hands-on segments. Troubleshooting together is part of the pedagogy.
- Providing a shared dataset vs allowing learners to bring their own imagery changes the time requirements significantly. Shared data is faster to set up. Learner data is more motivating.

Part 3: Lesson-by-Lesson Reference

This section is a quick reference for each lesson. Read it before class, not during. Each entry describes what the lesson does, what data or model it uses, the key teaching moment, and any common student difficulty.

IM-11.1 Chapter 1: Introduction to Marine Imaging

1.1 Introduction to Marine Imaging

Core lesson

This lesson sets the stage for the whole course. It covers why marine imagery exists, its historical arc from early underwater photography to modern ROV and AUV surveys, and why applying computer vision to ocean data is genuinely harder than applying it to terrestrial imagery. No code is involved. The key teaching moment is making water physics concrete: light attenuation, backscatter, and turbidity are not just background trivia. They directly affect which models work and which fail. Students often arrive assuming that a model trained on ImageNet will transfer cleanly to underwater footage, and this lesson is the right place to challenge that assumption.

1.2 Imagery Fundamentals (optional)

This lesson covers pixel bit depth, color spaces (RGB, HSV, grayscale), sensor types, and compression artifacts. It is most useful for students who have never worked with raw image data before. The key teaching moment is showing a JPEG compression artifact next to a PNG of the same image region. Students often have strong reactions when they see how much information is discarded, and that reaction is useful. There is no model training in this lesson.

1.3 Ocean Image Data

Core lesson

This lesson surveys the major sources of marine imagery: ROVs, AUVs, towed camera systems, aerial platforms, and satellites. Each source has a characteristic spatial resolution, coverage footprint, and set of deployment constraints. The key teaching moment is a mapping exercise where students match image source to resolution and coverage trade-off. No code is involved. Students who work through this lesson have a much clearer sense of why dataset choice matters later.

1.4 Introduction to AI

Core lesson

This lesson explains what a neural network is, how training works at a conceptual level, and what loss curves are telling you. The key teaching moment is showing a real training curve from the YOLO ice seal run. Students often conflate low training loss with good generalization. Addressing that early, with a real example, saves confusion later when their own models overfit. No hands-on training in this lesson, but students should read the curve carefully and describe what they see before the explanation is given.

1.5 Transfer Learning (optional)

This lesson explains why pretrained models work for marine imagery even though they were trained on terrestrial images. The key teaching moment is a visualization of which convolutional layers activate on a coral image when the model is a pretrained ResNet. Low-level layers respond to edges and textures regardless of domain. High-level layers are where the domain shift shows up. Students find this conceptually useful when they later have to decide whether to fine-tune all layers or freeze some.

1.6 Model Interpretability (optional)

This lesson introduces GradCAM and saliency maps. The worked example is the Whale Shark lesson. The key teaching moment is when the GradCAM heatmap highlights the background water rather than the organism. Students sometimes think this means the analysis failed. It does not. The model learned something about the water conditions associated with whale shark photographs, and that is itself a finding worth discussing. The lesson builds intuition for interpreting interpretability results honestly rather than selectively.

IM-11.2 Chapter 2: Introduction to Computer Vision

2.1 Image Annotation for CV

Core lesson

This lesson covers the three main annotation formats used in computer vision: bounding boxes, polygons, and keypoints. Students use CVAT or Label Studio to annotate a shared set of marine images. The key teaching moment is having two or more students annotate the same octopus image and then compare results. They rarely agree exactly on where the boundary is, and that disagreement is the lesson. Annotation quality drives model quality, and this experience makes that point more effectively than any explanation.

2.2 Manual Annotation (PIL/OpenCV)

Core lesson

Students draw bounding boxes and masks programmatically using PIL and OpenCV. The key teaching moment is showing the YOLO annotation format: a plain text file where each line is `class cx cy w h`. Students who have only used annotation GUIs often imagine something more complicated. Seeing that the output is just numbers in a text file demystifies the pipeline and makes later training steps feel less like a black box.

2.3 ImShow Data Exploration (optional)

This lesson covers methods for visualizing an image dataset before training. Students browse real annotation sets using interactive displays. The key teaching moment is almost always the same: students find at least one mislabeled or corrupt image in the dataset they are given. This is not staged. Real datasets have errors, and learning to find them before training is a practical skill. Budget time for the discussion that follows.

2.4 Image Manipulation with PIL and OpenCV (optional)

This lesson covers color space conversion, resizing, cropping, and histogram equalization. The key teaching moment is applying CLAHE to a dark deep-sea image. The improvement is immediate and visible, and students who find the earlier lessons abstract start to see the practical value of image preprocessing. No model training is involved.

2.5 Creating and Augmenting Datasets

Core lesson

Students use the Albumentations library to apply augmentation pipelines to marine images. Augmentations covered include flips, rotations, color jitter, blur, and cutout. The key teaching moment is applying an augmentation that is too aggressive for the deployment domain and observing that it can hurt model performance rather than help it. Underwater images have constraints: flipping a fish upside-down may be unrealistic, depending on the species. The lesson teaches students to think about whether an augmentation is plausible for their specific use case.

2.6 Augmentation Streamlit App (optional)

This is an interactive Streamlit application that lets users adjust augmentation parameters and see the effect on sample images in real time. It is particularly useful for workshops involving domain experts such as marine biologists who need to evaluate whether a proposed augmentation looks realistic for their imagery. No code-writing is required. The lesson takes under an hour.

2.7 Understanding CV Metrics

Core lesson

This lesson covers precision, recall, mAP50, mAP50-95, confusion matrices, and F1-confidence curves. It is one of the most important lessons in the course and often takes a full class period. The key teaching moment is showing that a model with 0.95 mAP50 on a validation set can still fail on imagery from a different deployment site, camera angle, or time of year. Out-of-distribution failure is not a bug in the metric. The metric is accurate; the distribution shifted. Students who understand this distinction leave with a much more honest view of what model evaluation actually tells you.

IM-11.3 Chapter 3: Computer Vision Development

3.1 Image Classification with Keras (optional)

This lesson fine-tunes a convolutional neural network using Keras to classify benthic animals, specifically crabs versus rockfish, from a dataset of approximately 800 images. The key teaching moment is observing the effect of class imbalance on the confusion matrix. When one class has far more training examples than the other, the model learns to favor it. The lesson makes this visible and then introduces simple remedies. Students who have only seen balanced toy datasets find this moment clarifying.

3.2 Introduction to YOLO

Core lesson

This lesson introduces the YOLO framework. It covers the four main tasks (detect, segment, classify, pose) and the four main modes (train, val, predict, export). The key teaching moment is running the default pretrained model on a marine image before any fine-tuning. Students see what a model trained on COCO detects when it looks at an ROV transect frame. The results are often funny and instructive at the same time. They motivate the need for fine-tuning and localization.

3.3 Training and Deploying Object Detection with YOLO

Core lesson

Students train a YOLO model on the Win2026TrashTank marine debris dataset. The lesson covers the three components of the YOLO detection loss: box loss, classification loss, and DFL loss. The key teaching moment is reading TensorBoard loss curves during training and explaining what each component means. Students learn to spot overfitting (diverging train and val curves), underfitting (both curves high and flat), and a healthy training run. This is the first time most students have watched a model train, and it tends to generate good questions.

3.4 *TrainandDeployObj ResNet (optional)*

This lesson covers the same object detection task using Faster R-CNN with a ResNet-50 FPN backbone via the PyTorch torchvision library. The key teaching moment is comparing the two-stage pipeline (Region Proposal Network followed by an ROI head) to YOLO's single-stage approach. Students learn that the architectural choice involves trade-offs in speed, accuracy, and ease of implementation, and that neither approach is universally better.

3.5 *TrainandDeployObj Keras (optional)*

This lesson repeats the detection task using KerasCV's RetinaNet. It is primarily useful for instructors whose students are already familiar with the TensorFlow ecosystem. The key teaching moment is Focal Loss and the intuition behind it: downweighting easy negatives so that the model spends most of its gradient on hard examples. This is particularly relevant for marine datasets where the background (open water, sand) vastly outnumbers the objects of interest.

3.6 *Ice Seal Classification with YOLO (optional)*

This lesson classifies four ice seal species from aerial survey imagery. It uses a transect folder structure to run batch inference across an entire survey grid. The key teaching moment is per-segment probability heatmaps that aggregate model predictions spatially. Students see how a computer vision model can produce a spatially distributed count that resembles what a field biologist would produce manually, and they can evaluate where the model is confident versus uncertain across the survey area.

3.7 *OceanCV Localizing YOLO*

Core lesson

This lesson uses the OceanCV FirstPass ROI detector as a starting point and implements a human-in-the-loop pipeline to adapt its generic detections to specific biological classes. The dataset is `transect_compressed.mp4`, already included in the course repository. The key teaching moment is the concept of pre-labeling: using an existing model to generate candidate annotations that a human then corrects, rather than annotating every frame from scratch. This typically reduces annotation time by 50 to 80 percent, and students find it immediately practical.

3.8 *Keypoint Detection with YOLO (optional)*

This lesson detects anatomical keypoints on black cod (sablefish) to estimate body length from imagery. The key teaching moment is the YOLO keypoint annotation format: `class cx cy w h kp1x kp1y kp1v ...`, where the visibility flag `v` can be 0 (not labeled), 1 (labeled but not visible), or 2 (labeled and visible). Students often find the visibility semantics confusing at first. Taking time to work through a few examples by hand before running the notebook saves debugging time later.

3.9 Instance Segmentation with YOLO

Core lesson

Students train a YOLO segmentation model on marine imagery with polygon annotations. The lesson covers how masks are represented internally, how to use mask area as a proxy for organism size, and how to interpret segmentation output in ecological terms. The key teaching moment is the comparison between a bounding box and a mask when organisms overlap. Bounding boxes overlap in ways that make individual organism identification ambiguous. Masks can separate touching or overlapping animals if the training data is good. This distinction is important for density estimation and behavior analysis.

3.10 K-means Segmentation (optional)

This lesson applies classical unsupervised segmentation using K-means color clustering to marine images. No training is involved. The key teaching moment is the fundamental limitation: K-means clusters pixels by color and has no concept of biological class. The clusters are meaningful only with human interpretation. Students come away understanding the difference between a segmentation algorithm and a model that knows what it is looking at, which is useful framing for the deep learning lessons that follow.

3.11 MarineInst Segmentation (optional)

This lesson uses the MarineInst model, which is built on the Segment Anything Model (SAM). It demonstrates zero-shot instance segmentation on marine imagery without any fine-tuning. The key teaching moment is performance on transparent and camouflaged marine organisms, which are genuinely hard cases. SAM sometimes handles them well and sometimes does not. Walking through both kinds of results, and discussing why, gives students a realistic picture of what foundation models can and cannot do.

3.12 SRGAN Super-Resolution (optional)

This lesson uses SRGAN implemented in TensorFlow to enhance low-resolution marine imagery. The key teaching moment is the gap between visual quality and downstream task performance. Super-resolved images often look significantly better to a human but do not always produce better detection results when passed through a trained model. That disconnect is worth spending time on, because it challenges the intuition that more visually appealing output is better output.

3.13 Sea Star GAN (optional)

Students train a DCGAN to synthesize sea star images. The key teaching moment is GAN instability: mode collapse (the generator learns to produce one or two plausible images rather than the full range) and discriminator dominance (the discriminator gets too good too fast and the generator stops learning). Both are common, and students who try to use GANs for data augmentation without understanding them

tend to end up with a biased dataset. This lesson gives them enough intuition to know when to be skeptical.

3.14 Whale Shark Interpretability (optional)

This lesson fine-tunes ResNet18 on Wild Me whale shark photo-ID data and then applies GradCAM to visualize what the model uses to make its predictions. The key teaching moment is when the GradCAM heatmap focuses on the background water or environmental context rather than the whale shark's spot pattern. That is not an error. The model found a real correlate of individual identity in the background, possibly because certain individuals were photographed in characteristic locations. This is a genuine scientific result and worth discussing as such.

3.15 Edge Deployment Optimization (optional)

This lesson covers model export to ONNX and TensorRT, INT8 quantization, and inference benchmarking on Jetson edge hardware. The key teaching moment is the three-way trade-off between latency, accuracy, and power budget. A model that meets accuracy targets on a GPU-enabled workstation may miss latency targets on a buoy-mounted camera system running on a battery. Students learn to think about deployment constraints as first-class requirements rather than afterthoughts.

3.16 Live Inference with PyQt (optional)

This lesson builds a real-time inference GUI using PyQt5 and OpenCV's `VideoCapture`. The key teaching moment is the QThread architecture that keeps the GUI responsive while the model runs on a background thread. Students who try to run inference on the main thread find that the interface freezes. The fix is not complicated, but it introduces students to the idea that software architecture matters for deployed systems, not just model accuracy.

3.17 Benthic Semantic Segmentation (optional)

This lesson fine-tunes SegFormer (using the `nvidia/mit-b0` pretrained weights) to classify every pixel in reef imagery into one of five categories: coral, algae, sand, rock, and fish or other. The model outputs percent cover statistics per image. The key teaching moment is the comparison between semantic segmentation and object detection for ecological metrics. Bounding boxes count individuals but say little about spatial distribution. Pixel-level classification gives area-based metrics like percent coral cover that are standard in benthic ecology but hard to produce at scale with manual methods.

IM-11.4 Chapter 4: Synthesis Project

4.1 Dataset Selection

Core lesson

Students explore LILA BC, FathomNet, Kaggle, and Hugging Face Datasets to find a marine image dataset suitable for their project. The key teaching moment is evaluating dataset quality before committing to training. Quality markers include annotation consistency, class distribution, licensing, and whether the deployment context of the data matches what the student wants to model. Students who skip this evaluation often spend several weeks training on a dataset that does not actually support their scientific question.

4.2 Hugging Face Organization Setup

Core lesson

Students join the [OceanCV Hugging Face Organization](#), upload a trained model, and learn how to manage repository access. Instructors setting up an organization for their own institution can follow the [Hugging Face Organizations Setup Guide](#). Public model hosting makes results reproducible and citable: a model hosted on Hugging Face can be downloaded and executed by anyone in one line of Python.

Hugging Face Hub & Student Project Highlights

Exemplary Student Projects on the OceanCV Hub:

- **Pythia's Oasis Methane Seep Megafauna Model**

https://huggingface.co/OceanCV/Pythias_Oasis

Student Attribution: Emily Pinneo

Data/Expedition Attribution: Pythia's Oasis Research Cruise (AT42-17, Dr. Deborah Kelley / NSF OCE 1658201)

Description: A YOLOv12 object detection model trained on deep-sea ROV dive footage (1,000m depth off the Oregon coast). Detects 16 benthic megafauna classes including *Actiniaria*, black coral, soft coral, rockfish, eelpout, hagfish, and *Scotoplanes*.

- **Monterey Bay Aquarium Open Sea Live Cam Model**

https://huggingface.co/OceanCV/Monterey_Bay_Aquarium_Open_Sea_Live_Cam_Object_Detection_Model

Student Attribution: Finley Charnyshou

Description: A multi-species object detection model trained on 1080p live stream footage from the Monterey Bay Aquarium Open Sea exhibit. Identifies and tracks 8 species including yellowfin tuna, scalloped hammerhead sharks, pelagic stingrays, green sea turtles, and dolphinfish.

- **European Green Crab Classifier**

https://huggingface.co/OceanCV/European_Green_Crab_Classifier

Student Attribution: Ryan Luvera

Description: A citizen science computer vision model trained on iNaturalist

and GBIF research-grade imagery. Classifies the invasive European green crab (*Carcinus maenas*) against 9 native Salish Sea species (including Dungeness crab, red rock crab, and purple shore crab) to support coastal invasion monitoring across Washington, Canada, and Alaska.

4.3 Model Card Writing

Core lesson

Students write a structured model card documenting the purpose, training data, evaluation metrics, intended use, and known limitations of their model. The key teaching moment is that a model card is not a marketing document. It should include failure modes and the conditions under which the model should not be trusted. A model card that lists only successes is incomplete and potentially misleading for downstream users.

4.4 Streamlit Apps

Core lesson

Students build a web interface for their trained model using Streamlit. The key teaching moment is that non-technical stakeholders, such as fisheries managers or conservation staff, can interact with a model through a Streamlit demo without touching any code. The interface is the bridge between technical work and practical use. Even a simple file-upload interface that returns a detection overlay is enough to make a model accessible to a much broader audience.

4.5 Gradio Apps

Core lesson

This lesson covers building a model interface using Gradio as an alternative to Streamlit. The key teaching moment is the `gr.Interface` one-liner, which gets a working demo running in under ten lines of code. The Hugging Face Spaces integration means the demo can be hosted publicly for free. For students who want to share their project quickly, Gradio is often faster than Streamlit for the first prototype.

IM-11.5 VLM Chapter: Vision-Language Models (all optional)

VLM.0 VLM Introduction

This lesson gives a conceptual overview of vision-language models. It covers the three components shared by most VLMs: an image encoder, a language model, and a cross-modal fusion mechanism. The key teaching moment is calibrating trust. VLMs can describe marine images in natural language and identify objects by name without fine-tuning, but they hallucinate. A model that says “this is a hammerhead shark”

with fluent confidence can be completely wrong. Students need a framework for knowing when to verify.

VLM.1 MLP Projection

This lesson covers the simplest fusion architecture: a two-layer MLP that projects image features from the vision encoder into the token embedding space of the language model. The key teaching moment is recognizing this as the core of the LLaVA architecture. The simplicity is part of the point. A very simple bridge between a pretrained vision encoder and a pretrained language model can produce surprisingly capable multimodal behavior.

VLM.2 Q-Former

This lesson covers the Q-Former module from BLIP-2. A fixed set of learned query tokens attends to image features via cross-attention, compressing the visual representation before it enters the language model. The key teaching moment is the efficiency argument: the Q-Former reduces thousands of visual tokens to 32 query representations, which makes the language model component much cheaper to run without losing most of the useful visual information.

VLM.3 Native Multimodal

This lesson covers models like Qwen2-VL that process image patches directly as tokens, without a separate projection stage. The key teaching moment is the trade-off relative to architectures with explicit fusion modules. Native multimodal models remove the information bottleneck and can be more capable on complex visual tasks, but they require more memory and compute because the full image token sequence passes through the language model directly.

VLM.4 Perceiver Resampler

This lesson covers the Perceiver Resampler from the Flamingo architecture. Like the Q-Former, it is a cross-attention resampler, but it is designed to handle variable-length image inputs gracefully. The key teaching moment is why this matters for marine imagery: high-resolution transect images and multi-camera setups produce very different numbers of image tokens, and a fixed-size projection bottleneck breaks when the input size changes. The Perceiver handles this naturally.

VLM.5 High-Resolution Grounding

This lesson covers coordinate grounding, where a VLM outputs bounding box coordinates as plain text. The lesson uses LILA BC imagery. The key teaching moment is the failure mode introduced by text-format outputs. Grounding outputs are strings like `[0.12, 0.34, 0.56, 0.78]`, which need a parser to extract usable coordinates. That parser can fail. The lesson builds the habit of validating structured outputs from language models rather than trusting that they are always well-formed.

A Note on the Textbook Itself

The OceanCV textbook is actively maintained and updated as tools and datasets evolve. Lesson numbers and chapter organization may shift between editions. The URL oceancv.org always points to the current version.